Online Appendix to
When Yield Curves Invert Together: Predicting Currency Carry Losses

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Abstract

This appendix gives the complete signal algorithm and timing conventions, derives the return accounting and compensation benchmark, reports supplementary estimates and adverse robustness, and documents two reproducible public-data exercises. The first reveals a complete 64-rule family for an independently sourced 10Y–3M curve proxy; the second studies synchronized delivered easing and an exploratory country-combination screen. The exercises are kept distinct from the baseline 10Y–2Y evidence.

Contents

A Signal construction and timing

This appendix gives the complete deterministic state transition used in the paper. It fixes the information set and separates fresh entry from confirmed release. It does not establish that the tenor, breadth threshold, and release length were selected independently of the return history.

A.1 Universe and information time

The nine non-dollar G10 currencies are the Australian dollar, Canadian dollar, Swiss franc, euro (Deutsche mark before 1999), British pound, Japanese yen, Norwegian krone, New Zealand dollar, and Swedish krona. The U.S. curve is excluded from the baseline count. For country i, define

$$q_{i,t} = y_{i,t}^{(10Y)} - y_{i,t}^{(2Y)}, \qquad I_{i,t} = \mathbf{1}\{q_{i,t} < 0\}.$$

All yields are end-of-month observations. A state measured at the end of month t may condition only returns realized during month $t+1$.

The return sample ends in 2026:02. Most money-market-rate series end in 2026:01 and therefore position the 2026:02 portfolio return under the $t \to t+1$ convention. The Australian, Canadian, and Swedish rate series extend further and are truncated to the common formation endpoint. The final return month does not use same-month rate information.

A.2 Curve-level transition rule

Each curve carries a live indicator $L_{i,t}$, a consecutive-steepening counter $k_{i,t}$, and an eligibility flag. A fresh inversion observed at $t-1$ is a crossing from $q_{i,t-2} \geq 0$ to $q_{i,t-1} < 0$; it makes the curve live at t. The update order is delayed entry, steepening-counter update for curves already live, release after the second consecutive increase, and finally cross-country aggregation.

Table 1: Curve-level state transition

Prior condition	Information used at t	End-of-month action
Eligible and not live at $t-1$	$q_{i,t-2} \geq 0$, $q_{i,t-1} < 0$	Set $L_{i,t} = 1$; initialize $k_{i,t} = 0$
Live	$q_{i,t} > q_{i,t-1}$	Increase $k_{i,t}$ by one
Live	$q_{i,t} \leq q_{i,t-1}$	Reset $k_{i,t} = 0$
Live; counter reaches two	Any current slope sign	Set $L_{i,t} = 0$ after the second increase
Released but still inverted	$q_{i,t} < 0$	Remain inactive and ineligible
Released and non-inverted	$q_{i,t} \geq 0$	Become eligible for a future fresh crossing
Live	Current or lagged slope missing	Reset $k_{i,t} = 0$; do not release
Not live	Either side of the lagged crossing missing	Do not enter

A curve can therefore be released while still inverted. It cannot re-enter until it first becomes nonnegative and then crosses below zero again. An interrupted or unobserved steepening sequence resets the counter. These rules make the state depend on the path of slopes, not only on their current signs. They also impose two timing steps: a crossing at $t-1$ makes the curve live at t, and the resulting state positions the return in $t+1$.

A.3 Cross-country aggregation and raw benchmark

After updating all curves, form

$$N_t = \sum_{i=1}^{9} L_{i,t}, \qquad S_t = \mathbf{1}\{N_t \geq 2\}.$$

The baseline state S_t is active in 92 of 458 return months and forms fifteen contiguous episodes. The corresponding current-inversion benchmark is

$$R_t = \mathbf{1}\left\{\sum_{i=1}^{9} I_{i,t} \geq 2\right\}.$$

The return in month $t+1$ is paired with S_t or, in the benchmark, R_t. The raw state asks whether fresh entry and confirmed release add information beyond the current sign of the curves.

Table 2: Incidence of the synchronized-inversion state across eras

Era	Sample months	Active months	Active share (%)
1988–94	84	33	39.3
1995–2000	72	19	26.4
2001–07	84	17	20.2
2008–09	24	6	25.0
2010–19	120	3	2.5
2020–21	24	3	12.5
2022–26	50	11	22.0

Notes: Counts reproduce the baseline era accounting. They describe when the state occurs, not whether its conditional-return coefficient differs across eras.

A.4 Initialization and audit outputs

At the sample start, every curve is initialized as inactive. A curve can enter only after both sides of a fresh crossing are observed. Missing values neither trigger entry nor count toward release; a missing comparison breaks consecutiveness and resets the confirmation counter. The recursion produces, for every country-month, the slope, raw inversion, fresh-entry flag, live status, steepening counter, release flag, and eligibility status. It then produces N_t, S_t, and episode start, end, and duration. These fields are the minimum audit trail needed to distinguish crossing month, live-state month, and return month and to reproduce the fifteen-episode count.

The reported real-time audit recomputes classifications at 435 truncated month-ends and finds no changes in prior signal, count, or own-state classifications. Recomputing the conditioning series and sorts at 492 formation dates produces no portfolio-membership mismatches. One additional return lag attenuates the carry-spot coefficient from -12.9 to -6.3 points and the S&P-beta coefficient from -7.5 to -4.7; forward-shift placebos remove the adverse association. These checks support the implemented information clock, although they are not a substitute for a fully versioned vintage archive.

B Accounting and mechanism details

This appendix separates return accounting, an exposure-ordering device, and a candidate compensation friction. The first is definitional. The latter two organize the evidence but are not structurally identified by the predictive regressions.

B.1 Currency and portfolio returns

Let $s_{i,t}$ be the log dollar price of currency i and let $d_{i,t} = (i_{i,t} - i_{US,t})/12$. The paper defines the one-month return proxy and carry portfolio as

$$\begin{aligned} rx_{i,t+1} &= d_{i,t} + \Delta s_{i,t+1}, \\ rx^{\text{carry}}_{t+1} &= \frac{1}{3}\sum_{i\in\mathcal{H}_t} rx_{i,t+1} - \frac{1}{3}\sum_{i\in\mathcal{F}_t} rx_{i,t+1} \equiv C_t + X_{t+1}. \end{aligned} \tag{1}$$

The high-rate set is $\mathcal{H}_t$ and the funding set is $\mathcal{F}_t$. The construction makes $C_t > 0$ in every sample month, so the portfolio loses only when $X_{t+1} < -C_t$. The return proxy combines a simple annualized differential divided by twelve with a log spot change and is therefore a first-order monthly approximation, not an exact executable forward payoff. Empirically, C_t averages 3.9 percent per year and has a minimum of 0.7 percent. The active-state spot-return difference is -12.85 percentage points and the total-return difference -11.31, so known income absorbs only part of the currency loss.

To organize the cross-section, write

$$\Delta s_{i,t+1} = -\lambda_i g_{t+1} + \eta_{i,t+1}, \tag{2}$$

where $g_{t+1} > 0$ is an adverse component shared across currencies, λ_i is a latent loading, and $\eta_{i,t+1}$ is idiosyncratic. A diversified long–short portfolio partly reduces idiosyncratic components; it retains the shared component when average λ_i differs between its two legs. The expanding-window equity beta $\widehat{\beta}_{i,t}$ is a predetermined proxy for λ_i, not the same primitive parameter.

B.2 Yield slopes and synchronized information

A long-minus-short yield spread combines differences in expected short-rate averages with differences in maturity-specific term premia (Campbell and Shiller, 1991; Hamilton and Kim, 2002). Consequently, one inversion is consistent with expected easing but does not reveal whether the news is domestic, global, expectations-driven, or a term-premium movement. Synchronization raises the plausibility of shared information because several markets cross a salient boundary, but it does not identify the shared shock.

The exact-one versus exact-two estimates give this idea empirical content. Exactly one live inversion is associated with carry spot and total coefficients of 8.33 and 8.78 percentage points per year. At exactly two, the coefficients are -15.72 and -15.13. The sign reversal indicates nonlinearity in cross-country breadth within the sample. It does not establish two as an optimal population threshold.

B.3 Delayed compensation as a candidate friction

Let A_{t+1} indicate an adverse event that reduces carry payoff by $J > 0$. Write

$$rx_{t+1}^{\text{carry}} = C_t - JA_{t+1} + \xi_{t+1}, \qquad \mathbb{P}_t(A_{t+1} = 1) = p_t, \qquad \mathbb{E}_t[\xi_{t+1}] = 0. \tag{3}$$

Then $\mathbb{E}_t[rx_{t+1}^{\text{carry}}] = C_t - p_t J$. For a tradable excess return in a frictionless benchmark, the Euler condition $\mathbb{E}_t[m_{t+1} rx_{t+1}^{\text{carry}}] = 0$ requires compensation when losses occur in high-marginal-utility states. Publicly visible risk therefore does not automatically imply a negative conditional mean.

The candidate friction places an information lag in compensation. Suppose

$$C_t = \bar{C}_t + \mu J \mathbb{E}_{t-1}[p_t], \qquad \mu \geq 1, \tag{4}$$

where $\bar{C}_t$ is ordinary carry income and the second component prices the current-period hazard using the earlier information set. Combining equations (3) and (4) gives

$$\mathbb{E}_t[rx_{t+1}^{\text{carry}}] = \bar{C}_t + J\left\{\mu \mathbb{E}_{t-1}[p_t] - p_t\right\}. \tag{5}$$

Hence

$$\mathbb{E}_t[rx_{t+1}^{\text{carry}}] < 0 \quad \Longleftrightarrow \quad p_t > \mu \mathbb{E}_{t-1}[p_t] + \frac{\bar{C}_t}{J}. \tag{6}$$

The inequality is the model's economic content. A rise in perceived risk is insufficient: the current expected loss must exceed both previously embedded event compensation and ordinary carry income. Fresh synchronized inversions may mark such a belief innovation, but the paper does not observe p_t or identify the adjustment friction.

B.4 Conditional uncovered interest parity

The supporting Fama regression estimates the appreciation slope separately by state:

$$\Delta s_{i,t+1} = a_i^{(s)} + \phi_s d_{i,t} + u_{i,t+1}, \qquad S_t = s, \quad s \in \{0, 1\}. \tag{7}$$

Under the paper's quote convention, uncovered interest parity implies $\phi_s = -1$. The pooled estimate is 0.50 outside the state and -1.33 inside it. The point estimates say that high-rate currencies depreciate in the state in which carry loses. The state interaction has a Newey–West t-statistic of -2.39 but an episode-bootstrap p-value of 0.340. The pooled panel contains many currency-months, whereas the common state changes across only fifteen episodes; the episode result is therefore the appropriate warning against treating -1.33 as a precise structural slope.

C Supplementary evidence and robustness

All exercises in this appendix use the same historical sample on which the state was assessed. They measure sensitivity and economic breadth; they do not create a frozen out-of-sample test.

C.1 Distribution and episode influence

Removing each portfolio's full-sample bottom-decile months leaves negative carry spot and total state effects. Because the deletion conditions on the realized outcome, it diagnoses tail concentration rather than estimating a preferred conditional mean.

Table 3: State effects after removing bottom-decile months

Portfolio component	Remaining signal months	Difference (%/yr)	Episode wild p
Carry spot	77	-7.24	0.006
Carry total	77	-5.50	0.024
High-beta spot	77	-3.72	0.084
High-beta total	78	-3.68	0.060

Notes: Each regression drops that return series' full-sample bottom-decile months from both states. The exercise asks whether a few crash observations mechanically generate the conditional-mean result; it does not address selection among alternative signal definitions. Wild-bootstrap p-values cluster by IYC episode.

The median and every reported lower quantile also move left. For carry spot returns, the monthly median changes from 0.35 percent outside the state to -0.76 percent inside, the 25th percentile from -0.93 to -1.98, and the 10th percentile from -2.35 to -3.94. This evidence is more informative about a broad distributional shift than the fact that all five worst months occur under the lagged state.

The beta-sorted portfolio exhibits a different distributional pattern. Its bottom-decile frequency is 16.3 percent in active months and 8.5 percent outside them ($p = 0.046$); at a fixed -4-percent monthly cutoff, the corresponding rates are 9.8 and 4.6 percent ($p = 0.033$). Annualized volatility is nearly unchanged, 7.94 versus 7.75 percent, with a variance ratio of 1.05 ($p = 0.80$), and reported loss severity conditional on crossing the tail cutoff is similar across states. These outcome-defined diagnostics describe a higher realized tail frequency, not an identified change in disaster probability.

Leave-one-episode-out carry estimates remain within roughly one-third of the full-sample coefficient. The high-beta effect remains statistically detectable under every one-episode deletion, with a largest reported bootstrap p-value of 0.006. These checks reduce dependence on a single episode but do not overcome the limited number of episodes.

C.2 Calendar stability

The carry-spot coefficient is negative in both calendar halves. The high-beta result is less stable and is concentrated after 2005. The split is descriptive because the rule was not frozen on the first half before evaluation on the second.

Table 4: The headline estimates in two halves of the sample

	1988:01–2004:12			2005:01–2026:02		
Series	b	$t_{episode}$	wild p	b	$t_{episode}$	wild p
Carry spot	-11.41	-2.54	0.037	-15.85	-1.99	0.010
Carry total	-10.20	-2.04	0.052	-14.76	-1.84	0.047
High-beta spot	-4.01	-1.12	0.287	-11.91	-4.07	0.010
High-beta total	-3.03	-0.80	0.448	-11.54	-3.85	0.011
Months		204			254	
Signal months		54			38	
Episodes		7			8	

Notes: Each column reports the state regression $y_t = a + b\,\mathbf{1}\{\text{IYC}\}_{t-1} + e_t$ estimated separately by calendar half. Coefficients are annualized percentage points; $t_{episode}$ clusters by IYC episode, and wild-bootstrap p-values use $B = 9{,}999$. This is a stability check, not a frozen out-of-sample test.

C.3 State-definition sensitivity

The reported grids vary breadth, release confirmation, and the short maturity. Separate exactly-one and exactly-two indicator regressions have coefficients with opposite signs; they are not a direct contrast because each indicator is compared with a different complement. At-least-three and at-least-four rules have fewer active months and later coverage. Releasing after one steepening month is noisier; requiring three retains the state longer. Rebuilding the state with a 10Y–3M rather than 10Y–2Y slope preserves the broad adverse ordering but changes episode timing.

The raw current-inversion state is active in 165 months, compared with 92 for the live state. Relative to the raw count, the live state improves tail concentration rather than absolute capture at every cutoff. The raw count contains 9 of the 23 worst-5-percent months and 19 of the 46 worst-10-percent months, versus 8 and 15 for the live state. Because the raw count is active in 36 percent of the sample rather than 20 percent, its concentration ratios are lower—1.09 versus 1.73 at the 5-percent cutoff and 1.15 versus 1.62 at the 10-percent cutoff. Only in the extreme five-month tail does the live state dominate absolute capture, 5 of 5 versus 2 of 5. The latch can retain losses that arrive after curves begin to re-steepen, but its extreme-tail concentration is partly mechanical because the historical crash chronology helped motivate persistence.

Breadth above two mainly measures episode maturity rather than a monotone risk dose. Months with at least three live curves have median episode age six months, compared with three months for exactly-two months, and eleven of fifteen episodes begin at exactly two. Conditional on the baseline state, a three-or-more indicator adds no reported information with or without episode-age controls ($p \geq 0.27$). The exact-two contrast is therefore an entry/configuration result, not evidence that each additional inversion raises loss risk.

A reciprocal-age version of the current-inversion count separates two timing margins. It has a high-beta spot coefficient of -14.87 ($p = 0.001$), larger than the baseline live state's -7.53, but a smaller carry coefficient of -10.43 ($p = 0.040$) than the baseline's -12.85. Under the full control set, the age-weighted high-beta coefficient remains -13.82 ($p = 0.008$), whereas its carry coefficient attenuates to -7.55 ($p = 0.084$). These in-sample comparisons suggest that exposure repricing is concentrated near fresh information, while the path-dependent release rule is more important for carry losses that continue into re-steepening.

Table 5: Measurement perturbations and exposure ordering

State construction	Carry spot	Wild p	High-beta spot	Wild p
10Y–2Y baseline	-12.85	0.001	-7.53	0.010
10Y–3M slope	-4.68	0.146	-5.25	0.011
U.S.-term-premium-adjusted 10Y–2Y	-5.42	0.107	-6.17	0.004

Notes: Coefficients are annualized percentage-point active-state differences. The 10Y–3M row rebuilds every curve state using a different short maturity. The term-premium row removes each national slope's fitted exposure to the U.S. ACM 10Y-minus-2Y slope premium before rebuilding the state. Both perturbations attenuate carry while retaining the beta-sorted exposure result.

Table 6: State-definition multiverse: reported and untested dimensions

Dimension	Reported neighborhood	Remaining uncertainty
Breadth	At least 1–4; exactly 1–2	Frequency-matched and country-combination rules
Release	One-, two-, and three-month confirmation	Other persistence and duration-matched placebo rules
Tenor	10Y–2Y baseline; 10Y–3M check	Other maturities and inversion cutoffs
Freshness	Fresh crossing versus current inversion	Alternative age decay and entry delay
Geography	Non-U.S. G10 baseline; U.S.-inclusion checks	Other developed-market sets and country combinations
Portfolio	Three-by-three carry and beta sorts	Other leg sizes, volatility scaling, feasible forwards
Timing	Month-end recursion and one extra lag	Vendor timestamps, asynchronous closes, vintage data

C.4 Bilateral cluster and own-curve specifications

The full country-level horse race is reported below. The cluster includes the focal country's curve in every row. These regressions ask whether the configuration remains informative after controlling for the currency's own state and slope variables; they do not show that other countries' curves alone predict the focal currency.

Table 7: Bilateral cluster and own-curve specifications

Currency	Own-state months	Own state alone b (t_{NW}) $[p]$	Cluster alone b (t_{NW}) $[p]$
AU	47	−8.76 (−1.65) [0.044]	−13.21 (−1.83) [0.015]
CA	40	+0.71 (+0.19) [0.858]	−4.61 (−1.33) [0.203]
CH	12	−20.12 (−1.91) [0.303]	−0.89 (−0.16) [0.867]
EU	25	+5.26 (+0.62) [0.589]	−1.19 (−0.20) [0.831]
GB	59	−3.84 (−0.59) [0.585]	−3.00 (−0.63) [0.542]
JP	22	+4.58 (+1.03) [0.298]	+2.35 (+0.58) [0.565]
NO	82	+4.53 (+0.84) [0.415]	−3.96 (−0.71) [0.520]
NZ	48	−11.42 (−1.33) [0.282]	−13.28 (−1.99) [0.011]
SE	13	−13.05 (−1.02) [0.497]	−4.94 (−0.76) [0.416]
Pooled (currency FE)	—	—	−4.74 (−2.34) [− − −]

Currency	Cluster, with own state b (t_{NW}) $[p]$	Own state, with cluster b (t_{NW}) $[p]$	Full-spec. cluster b (t_{NW}) $[p]$	Own-curve joint Wald p
AU	−14.18 (−1.38) [0.133]	+2.19 (+0.23) [0.859]	−12.92 (−1.31) [0.262]	0.408
CA	−6.01 (−1.45) [0.207]	+4.50 (+1.04) [0.356]	−5.55 (−1.46) [0.282]	0.801
CH	+1.28 (+0.21) [0.848]	−20.95 (−1.81) [0.272]	+1.15 (+0.18) [0.875]	0.115
EU	−3.40 (−0.51) [0.584]	+8.13 (+0.80) [0.552]	−4.00 (−0.61) [0.560]	0.488
GB	−2.00 (−0.45) [0.622]	−2.82 (−0.44) [0.668]	−0.44 (−0.10) [0.928]	0.486
JP	+1.68 (+0.38) [0.733]	+3.41 (+0.71) [0.413]	+6.37 (+1.32) [0.251]	0.006
NO	−6.85 (−1.15) [0.289]	+7.43 (+1.42) [0.221]	−8.17 (−1.23) [0.281]	0.233
NZ	−12.00 (−2.06) [0.011]	−3.22 (−0.41) [0.748]	−8.56 (−1.26) [0.220]	0.373
SE	−3.73 (−0.61) [0.526]	−10.58 (−0.92) [0.540]	−2.99 (−0.48) [0.662]	0.150
Pooled (currency FE)	−4.74 (−2.34) [− − −]	−0.01 (−0.01) [− − −]	—	—

Notes: Spot returns against the U.S. dollar, 1988:01–2026:02. Each compact cell reports the annualized coefficient, Newey–West (12) t-statistic in parentheses, and wild episode-bootstrap p-value in brackets. The full specification adds the currency's own 10Y–2Y slope, its one-month change, its slope relative to the U.S. curve, and the average G10 slope and change. The final column is the Newey–West Wald p-value for the joint null on that currency's own state, relative slope, and relative level conditional on the cluster. The pooled row reports currency fixed effects and no bootstrap inference. The cluster used in every row includes the focal country's curve. This table conditions on own-curve information; it is not a leave-one-country-out signal.

C.5 Exposure breadth and composition

The carry and beta portfolios have persistent cores rather than changing membership only when the state turns on. New Zealand occupies the high-rate leg in 85 percent of all months and 91 percent of state months; Australia in 71 and 73 percent. Japan and Switzerland remain the principal funding currencies. There is also rotation around the core: Norway's carry-long share falls from 62 to 34 percent in state months, while the pound's rises from 43 to 70 percent. The return result is therefore not created solely by state-month resorting, but portfolio composition is not fixed.

Table 8: Complete portfolio composition in the full sample and active state

Currency	Long share Full	Long share State	Funding share Full	Funding share State
Panel A: carry portfolio				
New Zealand	85	91	2	1
Australia	71	73	4	0
Norway	62	34	3	5
United Kingdom	43	70	4	0
Canada	20	21	23	14
Sweden	15	12	33	36
Euro area	4	0	51	47
Japan	0	0	83	97
Switzerland	0	0	96	100
Panel B: equity-beta portfolio				
Canada	100	100	0	0
Australia	97	92	0	0
New Zealand	77	76	0	0
Norway	26	32	0	0
Euro area	0	0	46	45
United Kingdom	0	0	52	47
Sweden	0	0	34	59
Japan	0	0	73	65
Switzerland	0	0	95	85

Notes: Entries are percentages of months in which each currency appears in the indicated three-currency leg. Full denotes all 458 portfolio months; State denotes the 92 active-state months. Column sums can differ slightly from 300 because the displayed shares are rounded to integer percentages.

The seven floating emerging-market currencies are not used to construct the G10 state. All seven bilateral state coefficients are negative; only the rand is individually detected at conventional levels (-14.68, $p = 0.015$), while the rupee is marginal (-7.98, $p = 0.061$) and the other five estimates are imprecise. Pooled spot and total effects are -14.37 and -13.35 percentage points per year, with episode-bootstrap p-values of 0.104 and 0.079. In the same EM panel, source-reported pooled spot coefficients are -4.54 for the raw current count and -3.92 for its six-month age filter. This ordering favors the live construction in sample, but all three designs use the same global dates and the live estimate itself is imprecise at the episode level.

The common-sample comparison separates international spot repricing from compensation.

Table 9: Spot repricing and compensation across currency universes, 1996–2026

Universe	Return leg	Inactive mean	Active mean	State difference	Episode wild p
G10	Spot	2.96	−12.79	−15.75	< 0.001
G10	Total	6.22	−8.67	−14.89	0.003
G10+EM	Spot	0.46	−12.51	−12.97	0.176
G10+EM	Total	11.68	2.51	−9.18	0.235

Notes: Annualized percentages over the common 1996:01–2026:02 sample. The G10+EM universe adds seven floating emerging-market currencies and applies the same three-by-three rate sort. Active-state income implied by total minus spot is approximately 4.12 percent for G10 and 15.02 percent for G10+EM. The nearly equal spot means but different total means show how predetermined carry income changes the payoff from similar currency repricing.

The combined sort is economically concentrated. Emerging-market currencies occupy 99 percent of carry-long slots and 93 percent of high-beta-long slots; the Brazilian real alone enters the carry-long leg in 93 percent of months. The comparison is therefore close to a persistent emerging-market exposure against the yen–franc–euro funding bloc, not a broadly rotating sixteen-currency portfolio.

Baseline exposure tests use expanding S&P 500 betas. An oil-beta sort adds little independent variation because its return is largely spanned by the carry and equity-beta portfolios.

C.6 Conditional UIP and macro timing

Appendix B gives the conditional Fama specification. The pooled slope is 0.50 outside the state and −1.33 inside, but the state interaction has episode-bootstrap $p = 0.340$. The point estimate is consistent with high-rate currencies depreciating during the state, while the interaction remains imprecise at the episode level.

The carry-spot response is front-loaded. In the baseline projection it falls about 1.1 percent in the first month ($p < 0.01$) and 2.4 percent cumulatively by month four ($p = 0.06$). Onset-only and lag-controlled designs reach roughly −3.7 to −4.2 percent around months four through six, followed by a partial rebound. Because horizons overlap and the state is persistent, these are descriptive timing associations rather than causal impulse responses.

The leading indicator declines by about 0.6 percent over six months, with similar magnitudes around episode onsets and under lag controls. Industrial production declines about 1.8 percent at six months in the baseline projection but becomes imprecise with the fullest controls. The average G10 money-market rate is approximately flat for the following year across the baseline, onset, and controlled projections. The leading-indicator response is persistent across these specifications; the industrial-production response and a uniform coordinated-easing path are not.

C.7 Benchmarks, controls, and term premia

The U.S. inversion, average G10 slope, and its change do not span the synchronized state. Lagged NFCI and VIX controls also leave the carry coefficient near its baseline magnitude.

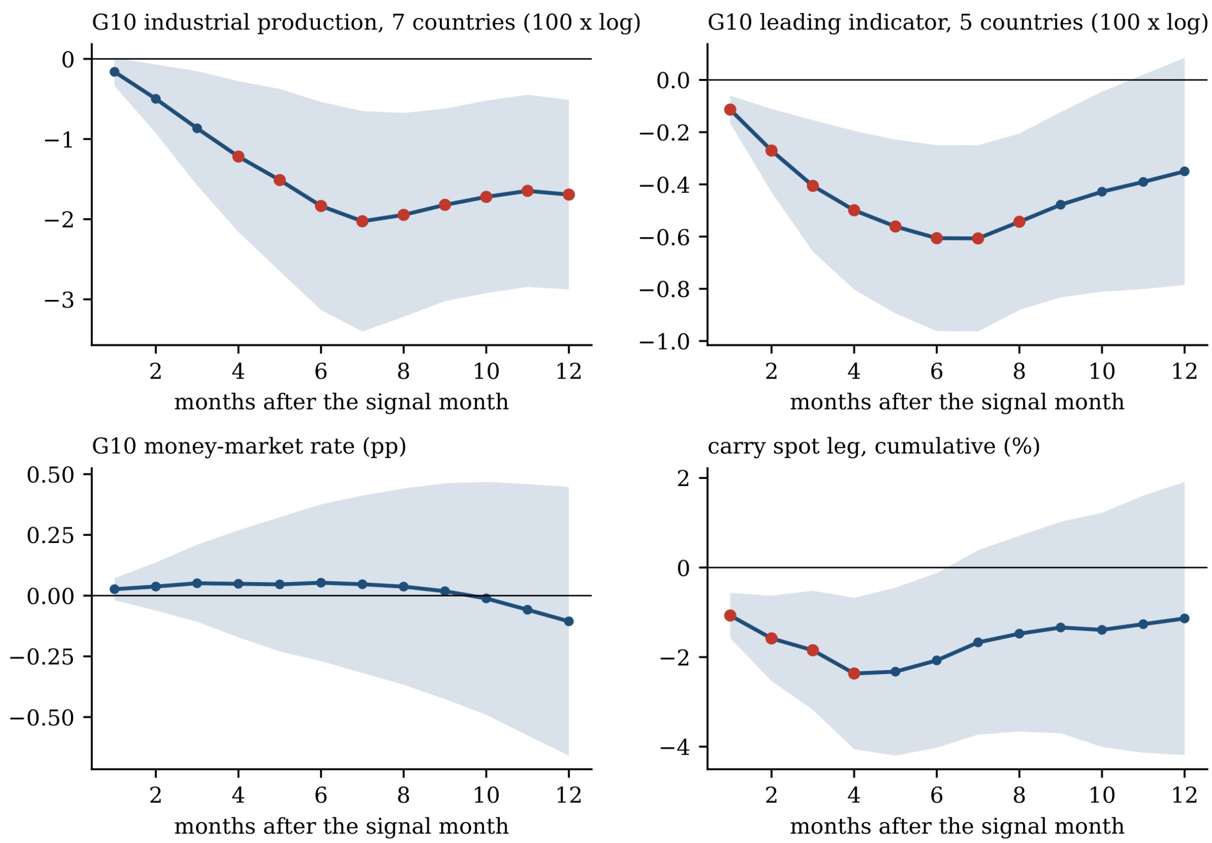

Figure 1: Outcomes following a synchronized-inversion state. The panels show local projections of cumulative changes in G10 industrial production, the OECD composite leading indicator, the average G10 money-market rate, and the carry spot leg. Bands are 90-percent Newey–West intervals; markers denote episode-bootstrap $p < 0.10$. Overlapping horizons and persistent states make these supporting temporal associations rather than causal responses.

Contemporaneous dollar and FX-volatility controls show that the realized loss is not simply the average dollar move or the measured volatility innovation, but they are not predictive controls.

Substituting G10 implied-volatility innovations preserves carry coefficients but weakens high-beta inference: high-beta point estimates remain near -7.5 percentage points and have p-values around 0.17–0.18. The raw current-inversion count is absorbed by the full NFCI/VIX set, with carry $p \geq 0.58$ and high-beta $p \geq 0.12$. These adverse cells prevent the control evidence from being read as uniform across volatility measures or state constructions.

The term-premium adjustment is asymmetric. After removing fitted exposure to the U.S. ACM 10Y-minus-2Y term-premium slope, 83 percent of national inversions remain. The rebuilt state still contains all five worst carry months, but it expands from 92 to 125 active months. The high-beta coefficient is -6.2 percentage points with $p = 0.004$, while the carry coefficient is about -5.4 with $p = 0.11$. The smaller carry estimate reflects a changed classifier—and possible dilution from the additional months—as well as removal of fitted exposure to the U.S. premium. The exercise does not identify what share of the baseline association is attributable to a common term-premium component. Comparable foreign-country decompositions are unavailable.

C.8 Interpretive boundary

The conditional return gap, distributional shift, and exposure ordering are statistically detectable in the reported specifications. The conditional-UIP interaction and industrial-production response are imprecise; emerging-market breadth and the downcurve split are exploratory. Feasible forward returns, complete data-vintage auditing, a comprehensive state multiverse, and a frozen future sample remain untested.

D Independent public-data checks

This appendix reports two executable exercises. The first constructs an independently sourced 10Y–3M yield-curve state and reveals its complete 64-rule family. The second examines outcomes around synchronized delivered policy easing. The yield proxy is a nearby measurement audit; the easing event occurs downstream of the predictive state. Neither is substituted for the baseline 10Y–2Y analysis.

D.1 A public 10Y–3M yield-curve proxy

D.1.1 Inputs, state, and outcome

The public curve panel uses current-vintage OECD monthly observations distributed through FRED. For each of the nine currencies, the slope is a 10-year government yield minus a 3-month interbank rate. The euro series uses Germany through December 1998 and the euro-area aggregate from January 1999. At least six observed curves are required to classify a month. Swiss and Japanese short-rate coverage begins later than the main sample, so the early state uses the available cross-section. Current vintages, the short-rate concept, the observation convention, and the euro splice all differ from the baseline end-of-month 10Y–2Y panel.

The live recursion uses the same timing sequence as Appendix A: a crossing observed at $t-1$ becomes live at t, a curve-specific counter records consecutive observed increases, release follows the required number of increases, and the country states are aggregated. A missing slope comparison resets the counter. The baseline-like public rule requires two live curves and two increases for release. The public outcome is a BIS policy-ranked log spot-return proxy, signed as foreign-currency appreciation against the dollar. For a one-month-lag rule, policy-rate differentials available by month t form the basket whose spot change is measured in $t+1$. For a two-month-lag rule, the basket is re-sorted using rates available by $t+1$ and its spot change is measured in $t+2$; it is not a month-t portfolio held for two months. The target is three currencies on each side. All currencies tied at a cutoff receive equal weight, so a realized leg can contain more than three currencies; an outcome is missing unless every selected nonzero-weight currency return is observed. The proxy is not a money-market or forward excess return.

Every country-month is retained in a curve-recursion ledger with the provider series identifiers, fresh-entry flag, live status, counter, release, eligibility, inversion, and observation status. A monthly ledger records observed curves, current and live counts, state, next-month outcome, onset, and episode. These files make the public state reproducible without discretionary event dating.

D.1.2 Declared specification family and inference

The declared family contains 64 rules. Live states cross four breadth thresholds, three release lengths, two return lags, and two portfolio leg sizes, giving 48 rules. Raw current-inversion states cross four thresholds, two lags, and two leg sizes, giving 16 more. The code reports every declared rule, but the declaration was not preregistered before the data were examined.

All family rules use the literal common complete-case calendar needed by both return horizons. For each rule, the reference distribution applies every common calendar displacement to the complete state sequence, includes the observed assignment, and doubles the smaller inclusive tail rank. The family reference standardizes each rule's rotation distribution and records the maximum absolute standardized coefficient across all 64 rules at each identical displacement. Under the maintained null that simultaneous cyclic shifts of the state sequences relative to returns are exchangeable, the resulting comparison controls the declared family. Without that assumption it is a finite common-calendar rotation-reference diagnostic, not an unconditional randomization probability.

Table 10: Independent public 10Y–3M proxy audit

Public-data specification	Log spot difference	Raw rotation ref.	Active months	Episodes
Baseline-like rule	-3.20	0.335	99	18
Exclude episodes containing 1998:09, 2008:10, or 2020:03	-0.46	0.826	88	15
First calendar half, 1988–2004	-5.73	0.382	49	9
Second calendar half, 2005–2025	-1.02	0.776	50	10
Delete five worst active months	$+0.94$	—	94	—

Notes: Annualized percentage-point active-minus-inactive differences in a BIS policy-ranked log spot-return proxy. Each leg targets three currencies; all currencies tied at a cutoff enter with equal weight, so realized leg size can exceed three. The baseline-like rule uses current-vintage OECD monthly 10-year government yields minus 3-month interbank rates distributed through FRED, fresh entry, breadth two, two consecutive increases for release, and a one-month return lag. It is distinct from the baseline 10Y–2Y state and is not an executable forward return. Across the complete declared family, 41 of 64 coefficients are negative, the baseline ranks 22nd from most negative to most positive, and the common-calendar max-$|z|$ rotation-reference value is 1.000. The crisis-deletion reference rotates the resulting complete-case sequence after excluded episodes are removed. The last row is outcome-conditioned and intentionally has neither a p-value nor an episode count, because deleting realized months mechanically splits state runs.

The baseline-like estimate is -3.20 percentage points per year, with raw rotation-reference value 0.335 and common-calendar maximum-standardized-coefficient reference value 1.000. Forty-one of 64 rules are negative; none meets the 5-percent family reference criterion. Figure 2 orders rules by active-state frequency and then their declared fields, not by the estimated return. The baseline-like rule ranks 22nd from most negative to most positive.

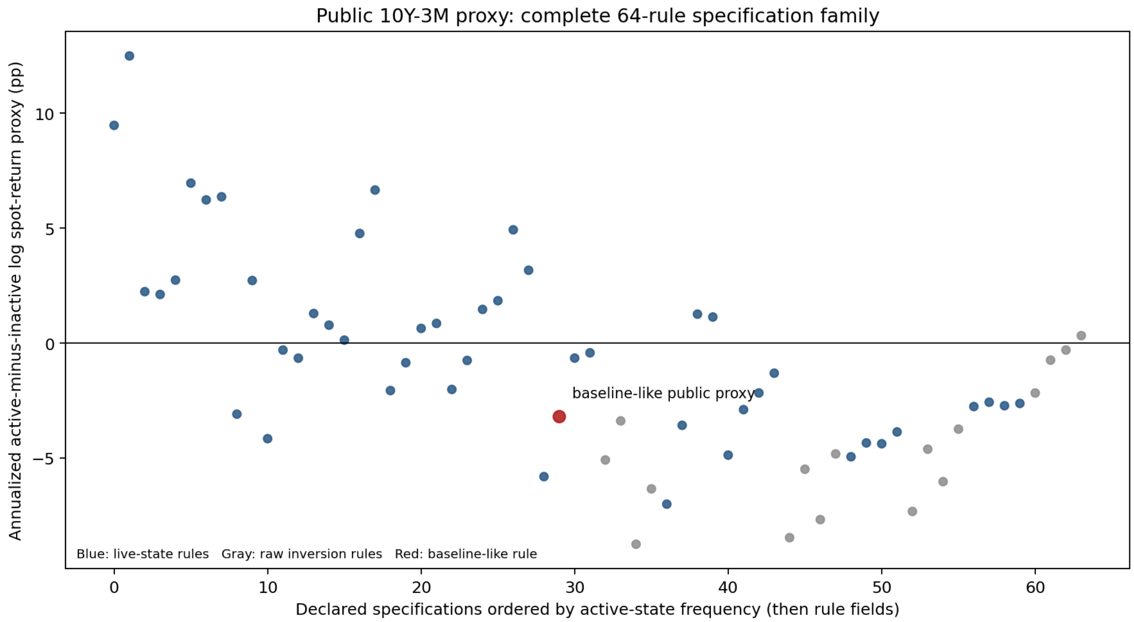

Figure 2: Complete public 10Y–3M rule family. Blue points are live states, gray points raw current-inversion states, and the red point is the baseline-like rule. Specifications are ordered by active-state frequency and fixed rule fields.

D.1.3 Exclusions, disjoint signals, and episodes

Removing each currency from both the public curve state and the outcome produces nine negative estimates. The samples and calendars overlap, so the signs are a sensitivity pattern rather than nine independent successes. For these influence and disjoint checks, the auxiliary reference rotates each complete-case sequence after dropping missing observations; it therefore compresses internal calendar gaps. The two geographically disjoint constructions produce different results: European curves precede a negative non-European basket difference, whereas non-European curves do not precede losses in the European basket.

Table 11: Public proxy exclusions and disjoint signals

Excluded from curve state and outcome	Log spot difference	Raw rotation ref.	Active months	Episodes
AUD	−5.08	0.133	69	16
CAD	−1.96	0.554	92	17
CHF	−4.51	0.204	92	17
EUR	−2.90	0.391	88	18
GBP	−0.60	0.827	88	16
JPY	−2.33	0.404	98	18
NOK	−1.32	0.655	70	14
NZD	−1.93	0.538	82	14
SEK	−1.95	0.591	91	16
Geographically disjoint curve state and outcome				
European curves → AUD/CAD/JPY/NZD	−15.84	0.062	38	10
Non-European curves → CHF/EUR/GBP/NOK/SEK	−0.34	0.887	44	9

Notes: Annualized active-minus-inactive differences in the log spot-return proxy. Each joint deletion removes a currency from both the public curve state and the public spot basket and requires five of the remaining eight curves. The resulting influence estimates overlap heavily and are not independent tests. Disjoint rows use no curve from a currency in the outcome basket. The European sensor requires four of five curves and targets one currency per side over four outcomes; the non-European sensor requires three of four curves and targets two per side over five. All cutoff ties are included and equally weighted. Reference values use inclusive circular shifts of each complete-case sequence after missing observations are dropped, compressing internal calendar gaps. They are not adjusted as an eleven-test family.

The public state produces eighteen episodes. Their cumulative outcomes are heterogeneous, with large negative contributions from the 1998 and 2008 episodes and several positive episodes. Removing episodes containing September 1998, October 2008, or March 2020 attenuates the baseline difference from −3.20 to −0.46 points. Deleting the five worst active months changes it to +0.94. The complete episode ledger makes this concentration inspectable.

Table 12: Public 10Y–3M proxy episode ledger

Episode	Onset	End	Months	Sum of log spot proxy	Worst month
1	1988:08	1989:12	17	−6.10	−4.62
2	1990:12	1991:05	6	−0.87	−1.15
3	1993:02	1993:02	1	−0.48	−0.48
4	1993:04	1993:05	2	−3.21	−3.05
5	1997:09	1997:09	1	−1.64	−1.64
6	1998:09	1999:01	5	−7.79	−7.04
7	1999:11	1999:12	2	+2.24	+0.26
8	2000:05	2001:02	10	+0.45	−2.31
9	2004:08	2005:07	12	+1.90	−1.61
10	2007:09	2008:04	8	−3.92	−3.66
11	2008:09	2008:10	2	−14.65	−11.66
12	2011:09	2011:12	4	+4.97	−2.47
13	2015:02	2015:05	4	−6.15	−4.03
14	2016:08	2016:09	2	+2.62	+0.96
15	2019:06	2019:11	6	−0.08	−2.95
16	2020:02	2020:05	4	−0.57	−6.08
17	2022:12	2023:09	10	+3.39	−1.59
18	2024:01	2024:03	3	+3.68	+0.34

Notes: Episodes use the baseline-like public 10Y–3M live state. The reported episode total is the sum of monthly percentage-point observations of the next-month BIS policy-ranked log spot-return proxy. Each side targets three currencies, with all cutoff ties included and equally weighted. The public 18-episode ledger is specific to this exercise and is distinct from the baseline 15-episode 10Y–2Y state.

The public audit produces an ambitious but bounded result. A nearby cross-country curve configuration yields negative coefficients in a majority of declared rules and in every joint delete-one influence check, yet the common-calendar family reference and crisis deletion remove conventional statistical detection. The exercise exposes measurement sensitivity using an independently executable current-vintage snapshot.

D.2 Delivered-easing mechanism proxy

The second exercise uses public data to examine outcomes around synchronized delivered policy easing. The event is intentionally later than the yield-curve state: inversions can encode expected easing, whereas policy-rate cuts record policymakers' actions. It tests selected mechanism implications but does not validate the inversion classifier.

D.2.1 Event construction

The sample is January 1988 through December 2025. Let $r^{pol}_{i,t}$ be country i's BIS end-of-period policy rate and define

$$K_{i,t} = \mathbf{1}\{\Delta r^{pol}_{i,t} \leq -0.10\}, \qquad B_t = \sum_i K_{i,t}, \tag{8}$$

where rates and the 10-basis-point cutoff are in percentage points. Let $D_t = 1$ when $B_t \geq 3$ and at least six country rates are observed. The onset indicator is

$$O_t = D_t \prod_{\ell=1}^{3} (1 - D_{t-\ell}). \tag{9}$$

Thus an onset requires at least three cutting countries after three months without another synchronized-cut month. The rule produces fifteen onsets. These fifteen public easing onsets are not the paper's fifteen inversion episodes; the identical counts are coincidental.

The public currency outcome is a spot-only proxy. BIS exchange rates are signed so that a positive return is foreign-currency appreciation against the dollar. At $t-1$, currencies are ranked by their policy-rate differentials against the United States; the proxy is long the three highest-ranked and short the three lowest-ranked. The rank is predetermined, but O_t is observed during month t, so month-t returns are contemporaneous event associations rather than feasible predictions made before easing.

Additional outcomes are CFTC non-commercial net positions scaled by open interest, the New York Fed ACM 10-year expected-rate and term-premium components (Adrian et al., 2013), the OECD G7 composite leading indicator, NFCI, and VIX. CFTC coverage is limited to seven directly matchable contracts and uses a documented Deutsche-mark/euro convention. ACM is a U.S. diagnostic, not a decomposition of the nine foreign slopes. The macro and financial-condition series are current vintage.

D.2.2 Event estimands and inference

The code uses two estimands. For a level outcome Y, the horizon-h event response is

$$\widehat{\theta}_h^{\text{change}} = \frac{1}{n_h} \sum_{j \in \mathcal{J}_h} \Big(Y_{\tau_j + h} - Y_{\tau_j - 1} \Big) . \tag{10}$$

For a monthly return flow R, it is

$$\widehat{\theta}_h^{\text{return}} = \frac{1}{n_h} \sum_{j \in \mathcal{J}_h} \sum_{\ell=0}^{h} R_{\tau_j + \ell}, \tag{11}$$

where τ_j indexes valid onset dates. Ninety-percent intervals resample onset episodes; leave-one-onset ranges diagnose influence. The reference distribution enumerates circular shifts of the complete onset sequence, retains shifts with the observed number of valid event outcomes, includes the observed assignment, and doubles the smaller inclusive-tail rank. With irregular missingness, the retained shifts need not form a transformation group; the resulting values are conditional finite-rotation references, not exact causal randomization tests.

Six declared primary outcomes receive Holm adjustment of their finite-rotation reference values: the shadow-carry spot return over months 0–1; the CFTC positioning change through month 3; ACM expected-rate and term-premium changes through month 3; the CLI change through month 6; and the NFCI change through month 3. The declaration is recorded in the analysis configuration and run manifest, but not in an immutable pre-analysis registration. VIX and other horizons are descriptive.

Table 13: Outcomes around synchronized delivered-easing onsets

Outcome	Estimate	90% CI	Rotation ref.	Holm adj.	Rotations
Policy-ranked shadow carry spot return, months 0–1	-1.06	[-3.47, 1.17]	0.222	1.000	441
Carry-aligned CFTC non-commercial net share, change through month 3	-6.42	[-16.34, 4.17]	0.568	1.000	81
ACM 10-year expected-rate component, change through month 3	-0.13	[-0.31, 0.07]	0.202	1.000	396
ACM 10-year term premium, change through month 3	-0.03	[-0.19, 0.14]	0.980	1.000	396
OECD G7 CLI, change through month 6	-0.34	[-0.72, 0.00]	0.159	0.952	353
Chicago Fed NFCI, change through month 3	0.03	[-0.08, 0.15]	0.520	1.000	396
VIX, change through month 1	1.89	[-2.30, 5.81]	0.400	–	20

Notes: Events are synchronized delivered-easing onsets constructed from public BIS policy rates. They are a downstream policy-rate state, distinct from the paper's predictive IYC state. Rotation-reference values enumerate every circular shift, condition on the observed number of valid episode outcomes, include the observed assignment, and double the smaller inclusive tail rank. Rotations reports the resulting finite reference count. The CFTC reference is small because missing contract coverage makes most rotations change the valid event count; its reference is correspondingly coarse. With irregular missingness, the same-count rotations need not form a transformation group, so these are conditional finite-rotation references rather than exact causal randomization tests. Intervals resample onset episodes. The Holm column adjusts the six declared primary reference values. Currency returns and CFTC positioning are percentage points, with CFTC positions measured as a share of open interest; ACM components are percentage points; CLI, NFCI, and VIX are index points.

The signs mostly fit a broad stress-and-easing chronology: the shadow carry proxy depreciates, carry-aligned positions decline, the expected-rate component falls, and the leading indicator weakens. Estimates are imprecise, and no primary outcome meets the 5-percent Holm-adjusted reference criterion. The U.S. term-premium estimate is nearly flat at three months. These results are directional mechanism evidence around policy delivery, not evidence that the inversion state predicts them.

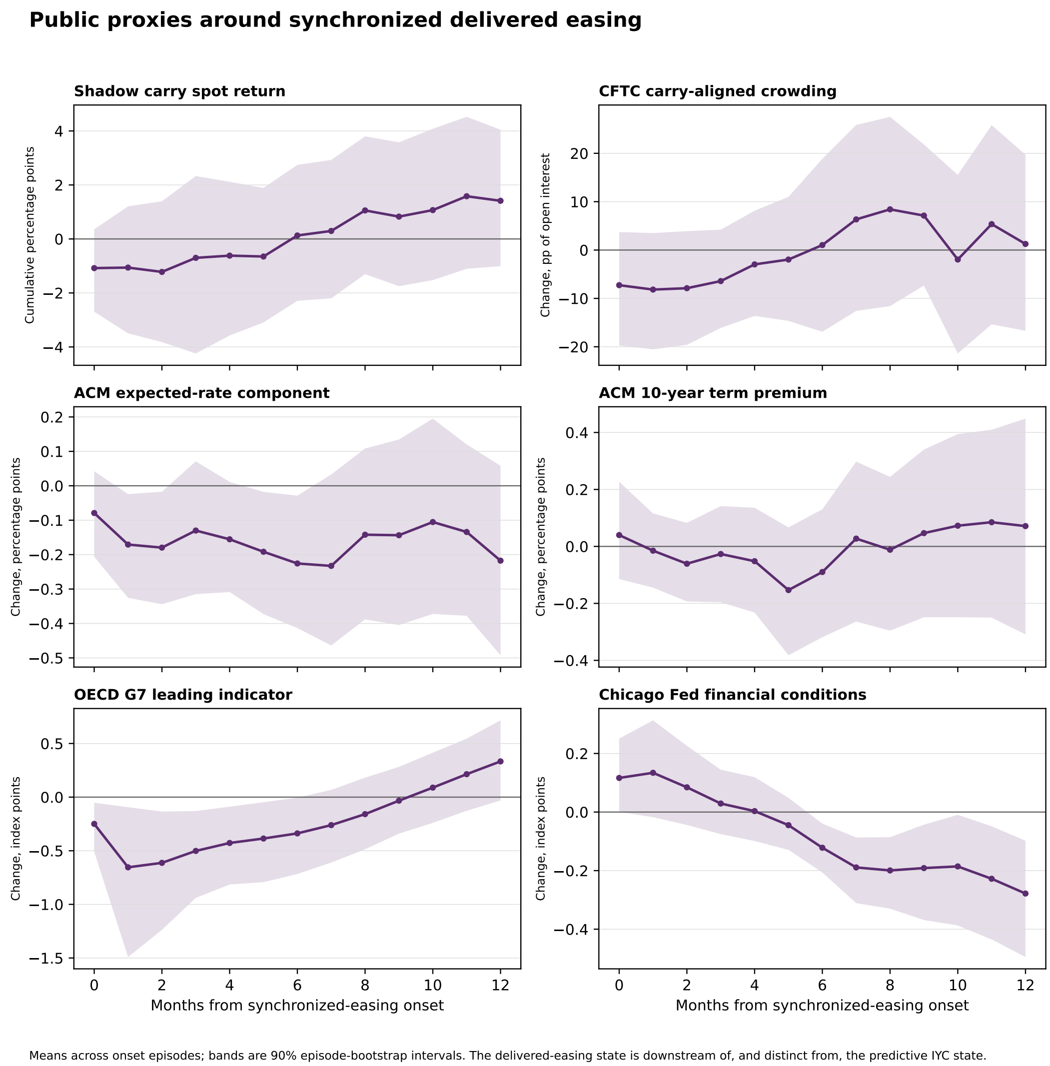

Figure 3: Outcomes around synchronized delivered-easing onsets. Event time zero is a month in which at least three policy rates fall by at least 10 basis points after a three-month quiet period. Panels show the code-defined cumulative return or endpoint-change estimands, with 90-percent episode-bootstrap intervals.

D.2.3 Sensitivity and interpretation

The shadow-carry response is recomputed using two- and four-country event thresholds and after omitting each event currency. Every threshold recomputes its own quiet period, so onset counts need not be monotone in the required number of cutting countries.

Table 14: Public-proxy sensitivity: shadow carry spot return

Construction	Estimate	Rotation ref.	Events	Rotations
At least three countries easing	-1.06	0.222	15	441
At least two countries easing	-1.10	0.120	22	434
At least four countries easing	-0.40	0.694	18	438
Exclude AUD from event rule	-0.56	0.515	17	439
Exclude CAD from event rule	0.35	0.548	18	438
Exclude CHF from event rule	-1.21	0.132	15	441
Exclude EUR from event rule	-1.02	0.182	16	440
Exclude GBP from event rule	-0.83	0.294	20	436
Exclude JPY from event rule	-0.24	0.838	19	437
Exclude NOK from event rule	-1.53	0.072	13	443
Exclude NZD from event rule	-0.53	0.590	15	441
Exclude SEK from event rule	-0.40	0.702	17	439

Notes: Each threshold definition recomputes its own three-month quiet window. Onset counts therefore need not be monotone in the number of countries required to cut. Leave-one-currency rows omit that currency from the event rule.

The exercise has four binding limits. Delivered easing is endogenous to prior information and occurs downstream of the paper's predictor. Policy-rate ranks are not money-market or forward rates, and spot returns omit carry income, basis, transaction costs, and implementation lags. CFTC, ACM, and current-vintage macro series are imperfect proxies for the intended objects. Finally, fifteen onsets—fewer for some outcomes—leave low episode-level power. The public results are therefore useful as a reproducible challenge to mechanism implications, not as a replication of the headline state.

E Which countries matter in the public proxy?

This appendix asks whether particular country pairs or triples dominate the public delivered-easing association. The public policy-rate exercise does not reveal which curves created the baseline fifteen inversion episodes. The analysis uses the same BIS policy-rate and exchange-rate data as Appendix D.

E.1 Combination screen

All 36 currency pairs and 84 triples are enumerated. For a given combination, an onset occurs when every member cuts its policy rate by at least 10 basis points after three months without that same joint cut. The outcome is the public shadow-carry spot return accumulated over event months zero and one. The ranking of all nine currencies uses lagged policy-rate differentials; the tested country combination defines the event, not the portfolio.

Reference values are reported only for combinations with at least six valid events. Every eligible event sequence is shifted through every circular calendar rotation. Raw two-sided rotation references are supplemented with a common-rotation maximum absolute standardized reference across every eligible pair and triple. Under the maintained null that simultaneous cyclic shifts are exchangeable, the maximum reference controls the declared family; otherwise it is a finite family diagnostic. The 120-combination family was declared in code and exhaustively reported but was not preregistered. Episode-resampling intervals, leave-one-event means, and alternative minimum-event rules diagnose small-sample fragility.

Table 15: Country combinations in the public delivered-easing proxy

Combination	Events	Mean $[0,1]$	Raw ref.	Max-$\lvert z\rvert$ ref.
CHF–GBP	6	-4.54	0.009	0.035
AUD–EUR	6	-4.26	0.004	0.055
GBP–SEK	9	-3.32	0.013	0.119
EUR–NZD	6	-2.80	0.053	0.486
CHF–SEK	7	-2.74	0.070	0.508
CAD–NOK–NZD	7	-4.09	0.013	0.066
GBP–NZD–SEK	8	-3.71	0.018	0.086
NOK–NZD–SEK	6	-3.62	0.026	0.235
CAD–GBP–NOK	6	-3.33	0.031	0.321
AUD–NOK–NZD	6	-2.87	0.048	0.519

Notes: The table reports the five most negative eligible pairs and triples. An event requires every listed country to cut its BIS policy rate by at least 10 basis points after three months without the same joint cut. The outcome is the cumulative policy-ranked log spot-return proxy in event months 0 and 1, in percentage points. Portfolio legs target three currencies per side and expand to include all cutoff ties with equal weight. Combinations require at least six valid events. Raw reference values use every circular timing rotation. The final column is the common-rotation maximum absolute standardized reference across every eligible pair and triple. Under simultaneous cyclic-shift exchangeability it controls the declared family; otherwise it is a finite family diagnostic. The family was not preregistered. This is a delivered-easing proxy, not a replication of the yield-curve state.

Of 120 combinations, 42 have at least six events. CHF–GBP is the only combination meeting the 5-percent common-rotation criterion: six joint-cut onsets have a mean cumulative two-month shadow-carry log spot-return proxy of -4.54 percentage points, raw reference 0.009, and maximum-$|z|$ reference 0.035. The 90-percent episode-bootstrap interval is $[-9.52, 0.31]$, however, and deleting October 2008 reduces the mean to -2.52. AUD–EUR is the next-most-negative pair, at -4.26 with maximum-$|z|$ reference 0.055. CAD–NOK–NZD is the most-negative triple, at -4.09 with reference 0.066.

The family result depends on admitting sparse combinations. With a six-event minimum, 42 combinations are eligible and one meets the 5-percent reference criterion. With minimums of eight or ten events, 25 and 15 combinations remain, no result meets that criterion, and the smallest maximum-$|z|$ reference values are 0.064 and 0.229. The evidence does not establish a stable country bloc.

E.2 Sensor countries versus loss-bearing currencies

The public lead does not match the baseline loss-bearing cross-section. CHF and GBP have near-zero or modest equity betas and small, imprecise state-conditioned currency losses. AUD and NZD bear the larger losses. The own-curve horse race also shows that a country's own curve generally does not forecast its currency after conditioning on the synchronized state.

This contrast suggests a testable distinction: countries whose bond or policy states reveal a shared episode need not be the currencies that bear its losses. Joint Swiss–U.K. easing may function as a calendar sensor for broad stress, while high-rate, high-beta currencies absorb the carry loss. The distinction is economically consistent with synchronization and exposure ordering, but the present result is only hypothesis-generating because delivered easing is downstream of forward-looking yield-curve inversions.

The direct test requires the nine-country live-state matrix. A fifteen-episode-by-nine-country contributor ledger could enumerate which curves activate each episode, then compare contributor combinations with the subsequent currency-loss cross-section under family-wide inference. The public combination screen specifies that future test but does not decompose the headline result.

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